Inspecting failures and anomalies

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

Prepare the purpose, prerequisites and materials

> Prepare Inspecting failures and anomalies: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Explain the **mechanism** and terminology of inspecting failures and anomalies

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Running the primary experiment

Without opening the old lesson, explain one key idea from Running the primary experiment. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of running the primary experiment

Reference example, not the only acceptable wording: Teacher explanation: The primary experiment follows the frozen plan and captures every eligible outcome. Unexpected failures are recorded rather than silently retried away.

Run declared cases, store raw paired results and deviations, and keep primary analysis separate from exploratory repairs. Stop or escalate only according to the stated rule.

Power and resource tradeoffs

Without opening the old lesson, explain one key idea from Power and resource tradeoffs. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of power and resource tradeoffs

Reference example, not the only acceptable wording: Teacher explanation: Power concerns detecting a declared effect under a specified model and test. Planning depends on effect size, variability, unit independence and budget.

Use supplied simulation results rather than an unsupported universal sample-size rule. Estimated detection frequency is detections/simulated alternative trials; false positives require null trials separately.

Privacy and data governance

Without opening the old lesson, explain one key idea from Privacy and data governance. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of privacy and data governance

Reference example, not the only acceptable wording: Teacher explanation: Data governance specifies who may use which records, for what purpose and for how long. Coded data can still be reidentifiable.

For fictional records separate participant-code table from analysis values; limit access and retention. Remove unnecessary fields and small-group details before a permitted aggregate export.

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. Failure inspection generates explanations to test. An unusual result is not automatically a bad data point. Use the supplied constructed scenarios to make the **mechanism** inspectable.

Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of inspecting failures and anomalies

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

Prerequisite answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

End goal for the pictorial case

Explain inspecting failures and anomalies using the records and boundaries of this fictional investigation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain and model the mechanism

> Model Inspecting failures and anomalies using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Failure inspection generates explanations to test. An unusual result is not automatically a bad data point.

Explain Inspecting failures and anomalies · 1

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Compare raw **output**s, **input** validity and procedure traces. Correct demonstrable measurement errors with provenance; retain valid unusual **observation**s. Label post-result investigations exploratory.

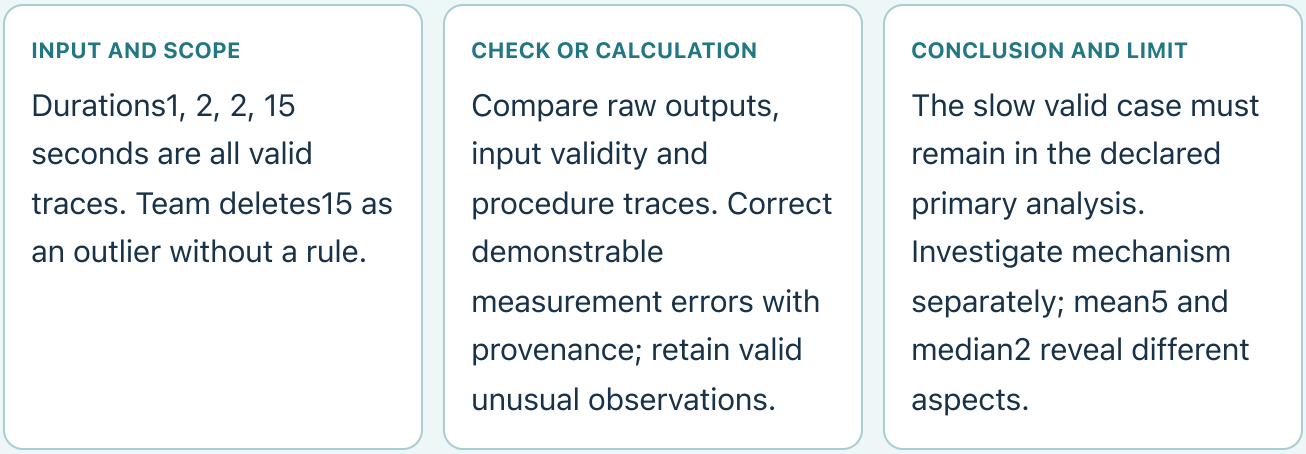
Explain Inspecting failures and anomalies · 2

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Anomaly	An unusual observation requiring investigation
Provenance-backed correction	A repair justified by source evidence
Exploratory analysis	A follow-up chosen after results

Inspecting failures and anomalies — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Explain Inspecting failures and anomalies · 3

Guided: The slow valid case must remain in the declared primary analysis. Investigate **mechanism** separately; mean5 and median2 reveal different aspects.

Qualification: Post-hoc exclusion based on an unwanted outcome can distort the conclusion.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Durations1, 2, 2, 15 seconds are all valid traces. Team deletes15 as an outlier without a rule.

The slow valid case must remain in the declared primary analysis. Investigate **mechanism** separately; mean5 and median2 reveal different aspects.

Model the reasoning aloud · 4

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

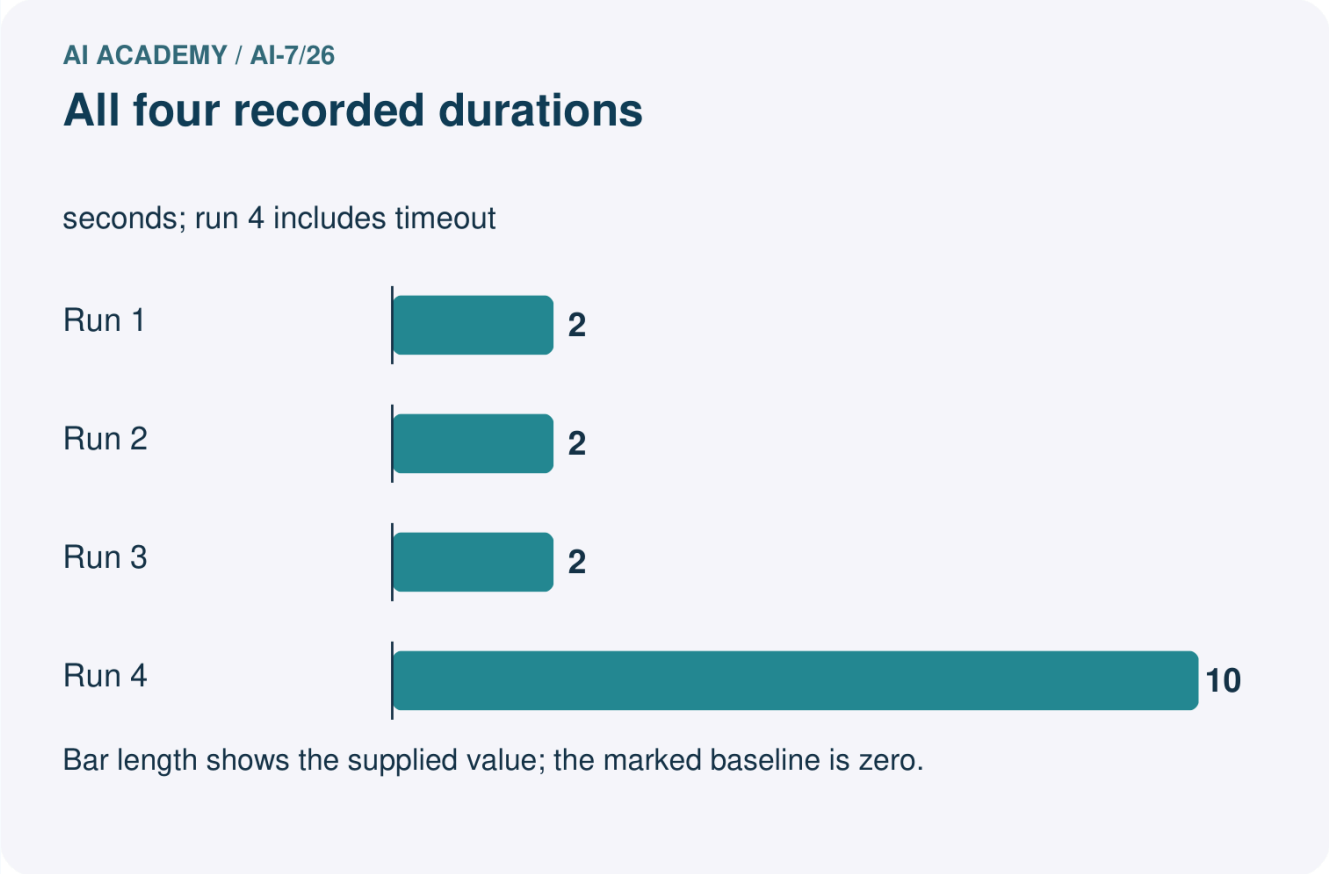
A pictorial worked case: Inspecting failures and anomalies

Inspect an **anomaly** without erasing it. A timing study records four runs and notices one unusually long duration.

The facts we will use

Supplied facts

Durations are 2,2,2,10 seconds. Logs show a source timeout on the fourth run. The plan includes timeouts in reporting.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/26

Inspecting failures and anomalies

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep the raw record

The fourth duration remains 10 seconds.



2. Compute summaries

Mean is 16/4=4 seconds; median is 2 seconds.



3. Investigate the event

Report the timeout evidence and distinguish it from an untested explanation.

CHANGE ONE THING

Delete the 10-second record only because it looks bad.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Keep the raw record	The fourth duration remains 10 seconds.
2. Compute summaries	Mean is 16/4=4 seconds; median is 2 seconds.
3. Investigate the event	Report the timeout evidence and distinguish it from an untested explanation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

An **anomaly** invites investigation, not automatic deletion. A slow failure may be part of real service behaviour; the predefined rule determines inclusion.

What this example does not establish

Keep the stated limit

A logged timeout does not identify every underlying cause; report conditions and test explanations.

Change one thing in the pictured case

Change one thing

Change: Delete the 10-second record only because it looks bad.

Predicted result

Prediction: The mean becomes 2 seconds but the selective report violates this plan.

Why the result follows

Reason: Observed inconvenient outcomes are not an approved exclusion reason.

Ask while showing the picture

Ask while showing the picture · 5

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Inspecting failures and anomalies, reduce support deliberately and check individual reasoning before discussing answers.

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Reduce help deliberately

Completion practice

Completion task: Durations1, 2, 2, 15 seconds are all valid traces. Team deletes15 as an outlier without a rule. Show the intermediate steps and give the conclusion with its scope.

Reasoning to listen for

Teacher check: Teacher explanation: The slow valid case must remain in the declared primary analysis. Investigate **mechanism** separately; mean5 and median2 reveal different aspects.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Inspecting failures and anomalies. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Post-hoc exclusion based on an unwanted outcome can distort the conclusion.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Calculate mean and median for 2,2,2,10. What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?

Reasoned teacher answer

The fourth duration remains 10 seconds. Mean is 16/4=4 seconds; median is 2 seconds. Report the timeout **evidence** and distinguish it from an untested explanation.

Changed case: The mean becomes 2 seconds but the selective report violates this plan. Observed inconvenient outcomes are not an approved exclusion reason.

Changed-case reasoning: The mean becomes 2 seconds but the selective report violates this plan. Observed inconvenient outcomes are not an approved exclusion reason.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

TEACHER GUIDE · 4 OF 9

Run the investigation and diagnose misconceptions

> Run the Inspecting failures and anomalies investigation and use the first unsupported step to choose a teaching response.

The worked result establishes every future case: Post-hoc exclusion based on an unwanted outcome can distort the conclusion.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Inspecting failures and anomalies questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Guided and exit questions

Guided: The slow valid case must remain in the declared primary analysis. Investigate **mechanism** separately; mean5 and median2 reveal different aspects.

Qualification: Post-hoc exclusion based on an unwanted outcome can distort the conclusion.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

Workbook question · W1

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher reasoning · W1

Teacher explanation: Failure inspection generates explanations to test. An unusual result is not automatically a bad data point. Compare raw **output**s, **input** validity and procedure traces.

Correct demonstrable measurement errors with provenance; retain valid unusual **observation**s. Label post-result investigations exploratory.

W2 Trace the worked case

Workbook question · W2

Durations1, 2, 2, 15 seconds are all valid traces. Team deletes15 as an outlier without a rule. Show the intermediate steps and give the conclusion with its scope.

Teacher reasoning · W2

Teacher explanation: The slow valid case must remain in the declared primary analysis. Investigate **mechanism** separately; mean5 and median2 reveal different aspects.

W3 Audit the limit

Workbook question · W3

A report omits this qualification: Post-hoc exclusion based on an unwanted outcome can distort the conclusion. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher reasoning · W3

Teacher explanation: Post-hoc exclusion based on an unwanted outcome can distort the conclusion.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W4 Apply a new case independently

Workbook question · W4

Values2, 3, 4, 20 are valid. Compute mean and median and propose an honest investigation.

Teacher reasoning · W4

Teacher explanation: Mean7.25, median3.5. Keep20; inspect its case/trace for a **mechanism** rather than silently deleting it.

W5 Design a revealing follow-up

Workbook question · W5

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher reasoning · W5

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Post-hoc exclusion based on an unwanted outcome can distort the conclusion.

The new case should test this particular **mechanism** rather than merely restate the original result.

Independent assessment and fresh retry

Teacher key · Independent assessment and fresh retry

W4: Mean7.25, median3.5. Keep20; inspect its case/trace for a **mechanism** rather than silently deleting it. Require intermediate reasoning and the stated scope. Fresh retry: One duration recorded2000ms among values in seconds.

Expected reasoning: Convert documented2000ms to2 seconds with provenance; this is a unit correction, not rejection because it looks large.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Independent question · withhold the key

Selected case: Values2, 3, 4, 20 are valid. Compute mean and median and propose an honest investigation.

Selected case answer

> **Teacher key · Selected case answer**
>
> Teacher explanation: Mean7.25, median3.5. Keep20; inspect its case/trace for a **mechanism** rather than silently deleting it.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

> **Teacher key · Selected case answer**
>
> Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.
>
> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect the project, support and next lesson

> Connect Inspecting failures and anomalies to Reproducible AI Research Study, adapt support and plan a later changed-case check.

Add a labelled inspecting failures and anomalies **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of inspecting failures and anomalies.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Robustness across conditions. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-7/27, ai-7/29, ai-7/32. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	

Record	Teacher evidence
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Inspecting failures and anomalies lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Explain the **mechanism** and terminology of inspecting failures and anomalies

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: Durations1, 2, 2, 15 seconds are all valid traces. Team deletes15 as an outlier without a rule. Show the intermediate steps and give the conclusion with its scope.

Supported practice answer

Reasoned practice answer: Teacher explanation: The slow valid case must remain in the declared primary analysis. Investigate **mechanism** separately; mean5 and median2 reveal different aspects.

Check every learner before discussion: Use the worked case for Inspecting failures and anomalies. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Post-hoc exclusion based on an unwanted outcome can distort the conclusion.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Delete the 10-second record only because it looks bad. Require a **prediction** and a reason before comparing. Expected result: The mean becomes 2 seconds but the selective report violates this plan.

Reason: Observed inconvenient outcomes are not an approved exclusion reason.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

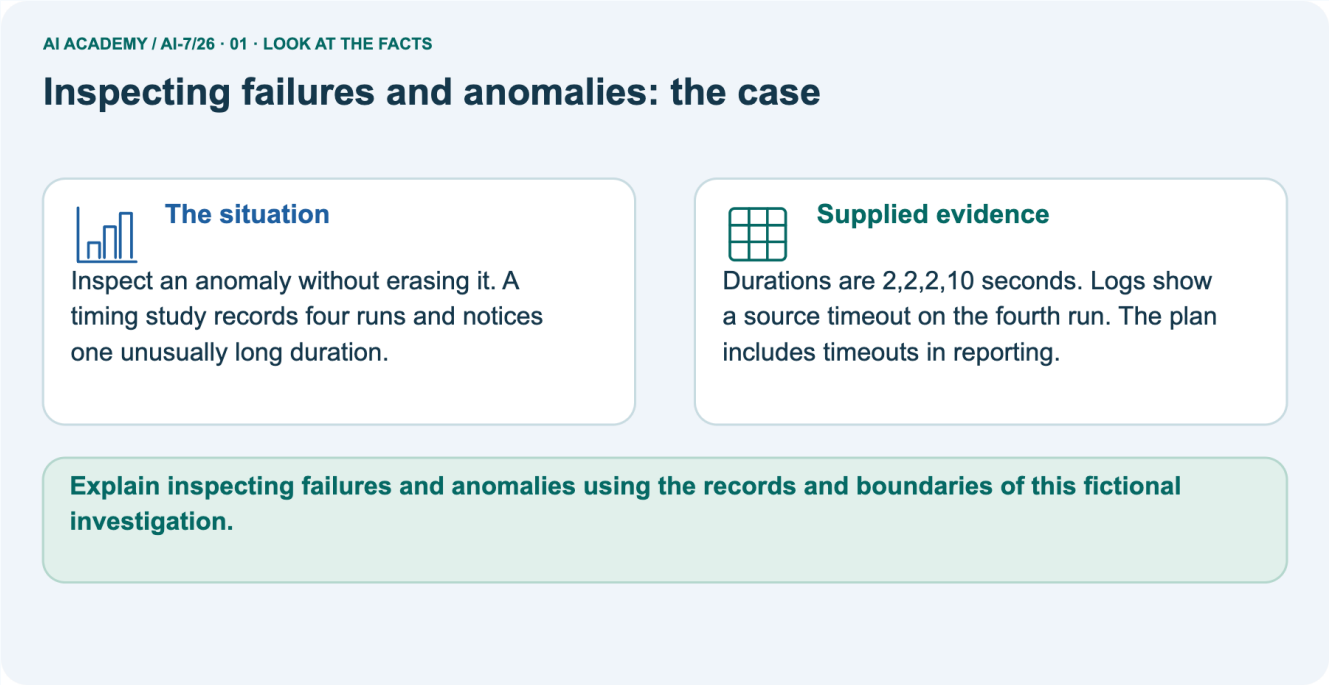
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Inspecting failures and anomalies graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Inspecting failures and anomalies: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Inspect an **anomaly** without erasing it. A timing study records four runs and notices one unusually long duration. Supplied facts: Durations are 2,2,2,10 seconds. Logs show a source timeout on the fourth run.

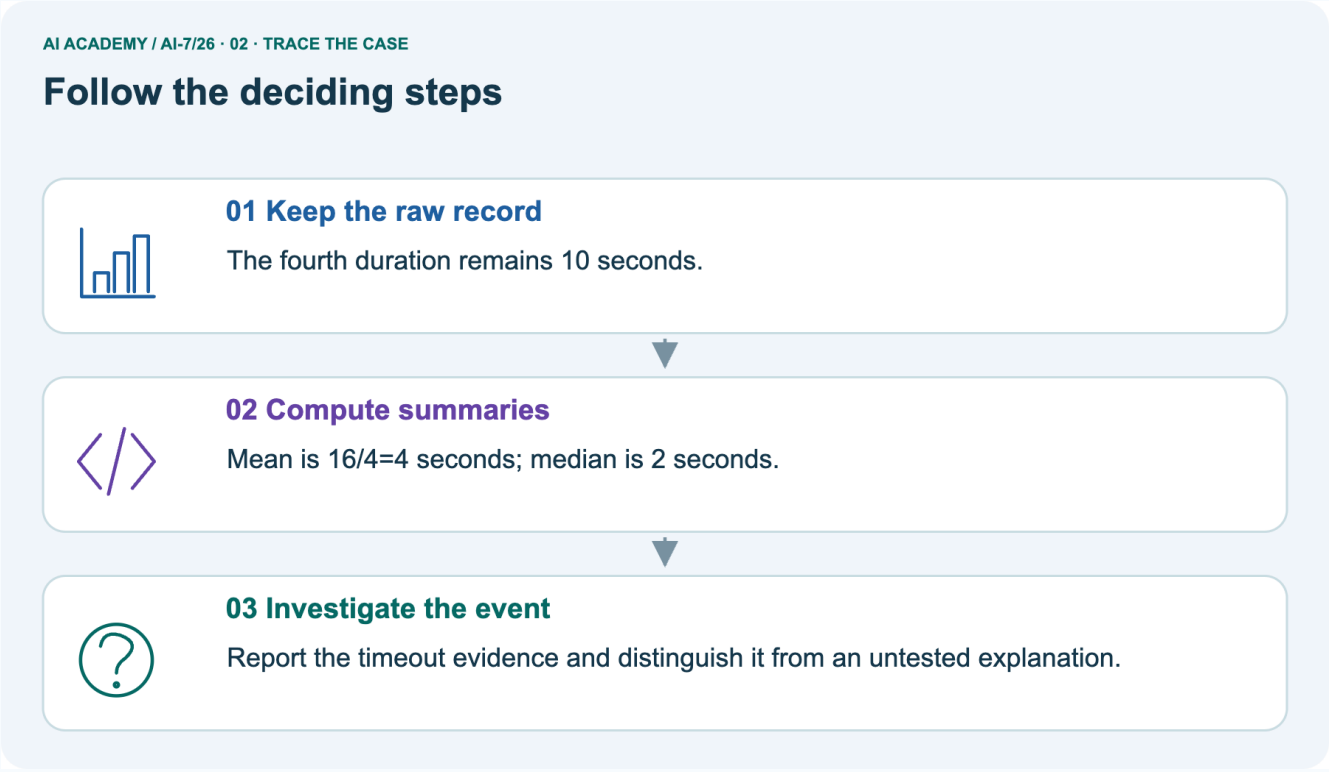
The plan includes timeouts in reporting.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

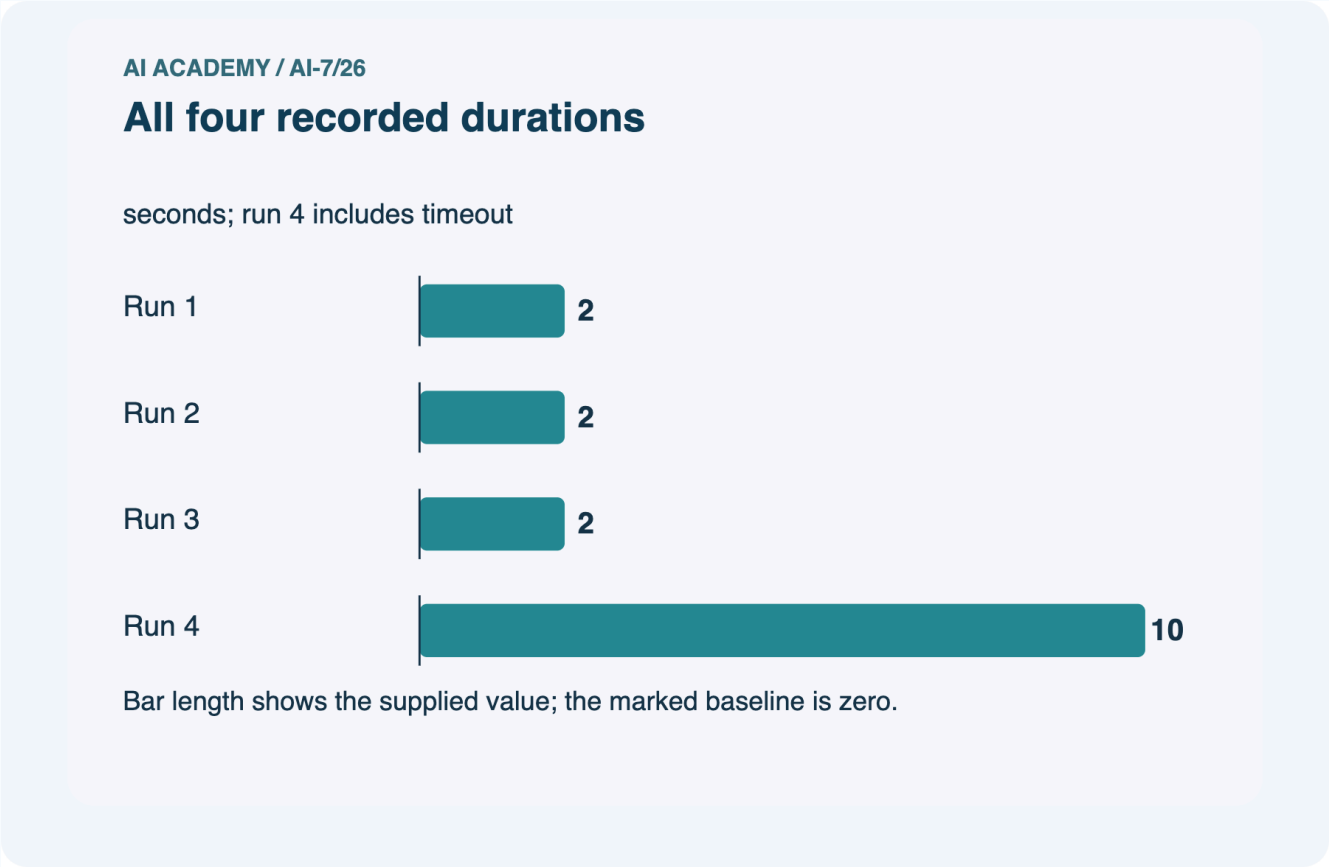
Keep the raw record: The fourth duration remains 10 seconds. → Compute summaries: Mean is 16/4=4 seconds; median is 2 seconds. → Investigate the event: Report the timeout **evidence** and distinguish it from an untested explanation.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. All four recorded durations



Follow the labelled picture with the matching explanation. Open the vector diagram

All four recorded durations. Units: seconds; run 4 includes timeout. Run 1: 2. Run 2: 2. Run 3: 2. Run 4: 10.

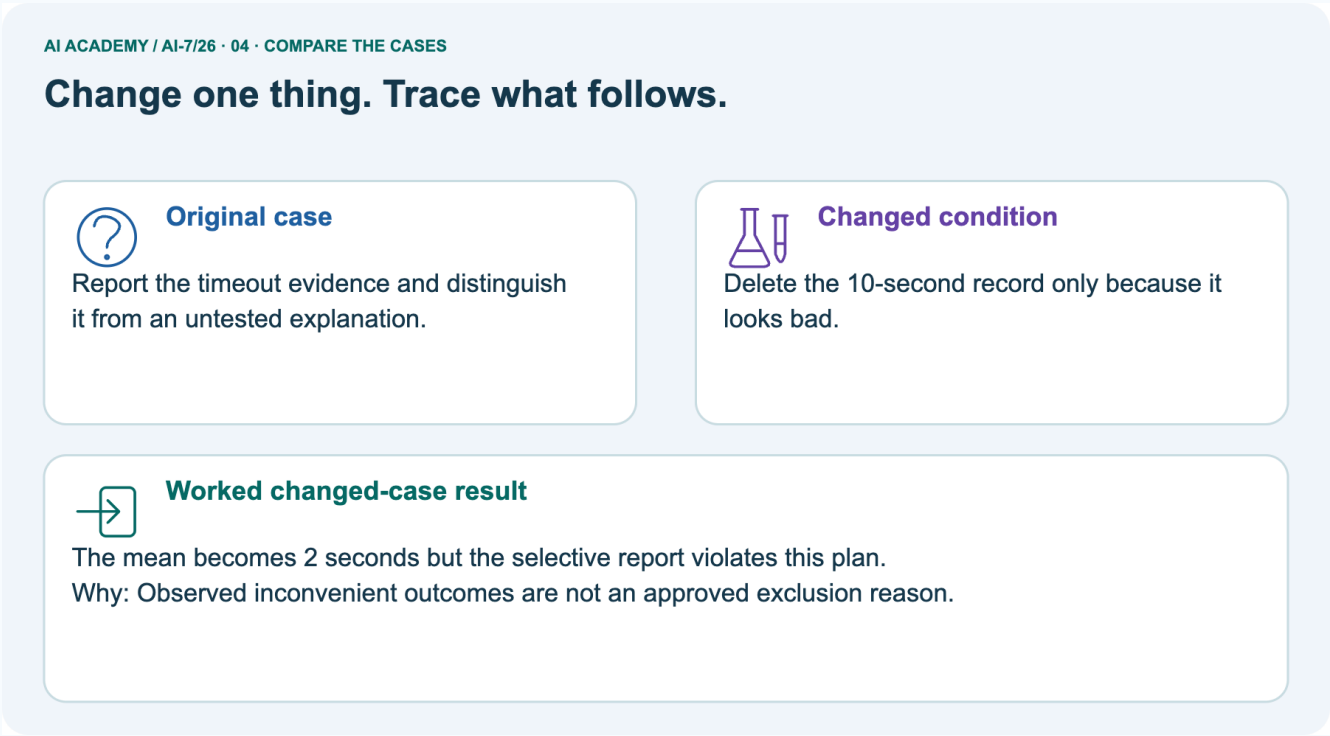
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Report the timeout **evidence** and distinguish it from an untested explanation.
Changed condition: Delete the 10-second record only because it looks bad.

Expected result: The mean becomes 2 seconds but the selective report violates this plan. Reason: Observed inconvenient outcomes are not an approved exclusion reason.

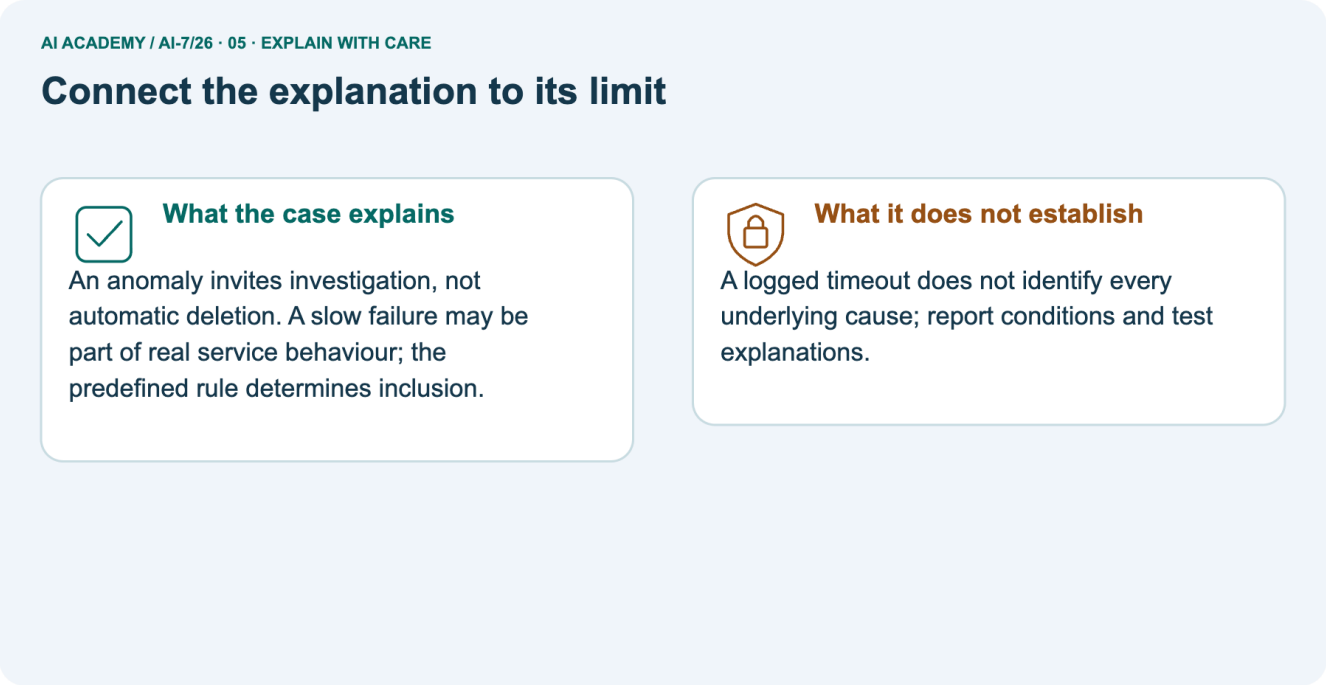
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Observed inconvenient outcomes are not an approved exclusion reason.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An **anomaly** invites investigation, not automatic deletion. A slow failure may be part of real service behaviour; the predefined rule determines inclusion.

Limit: A logged timeout does not identify every underlying cause; report conditions and test explanations.

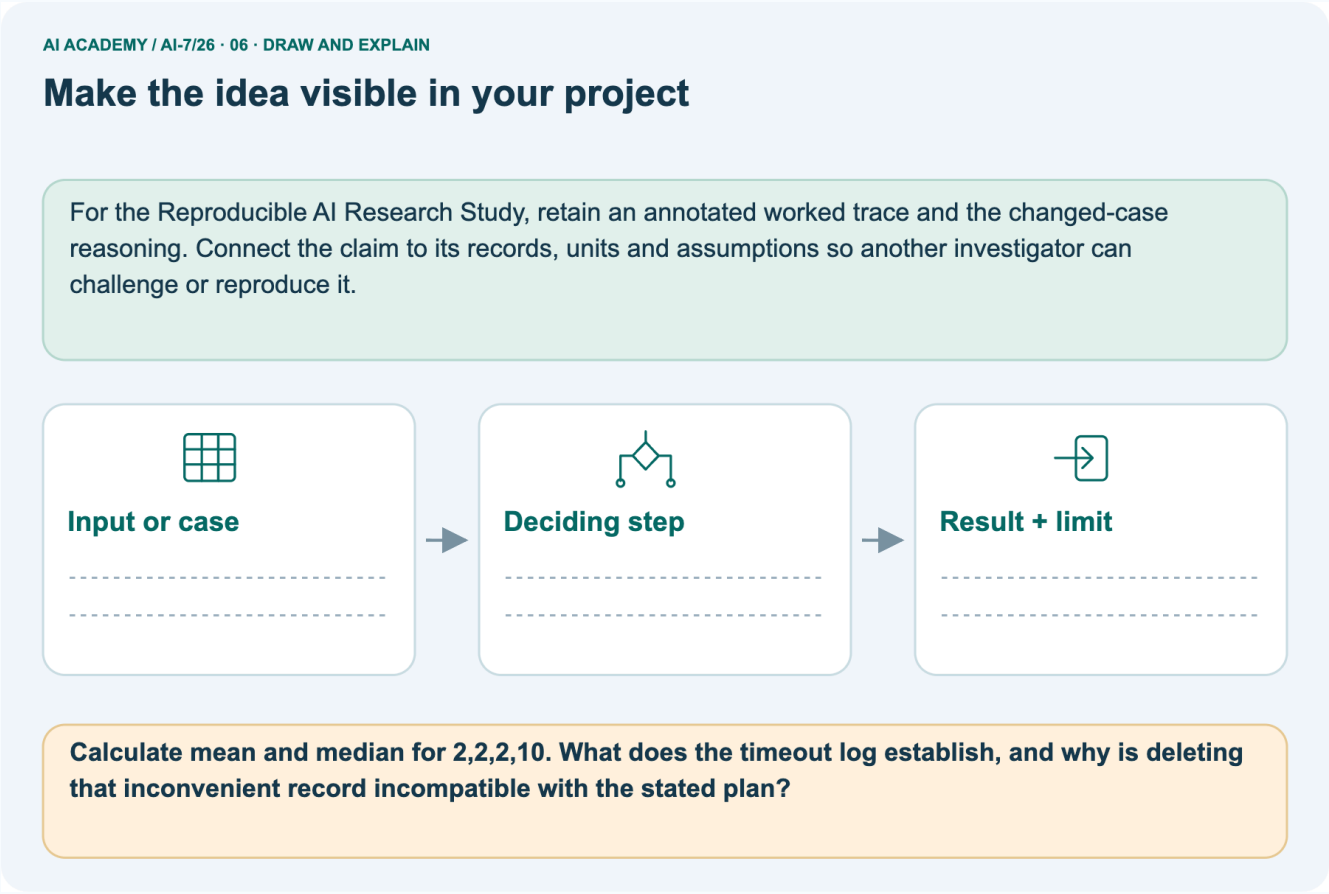
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A logged timeout does not identify every underlying cause; report conditions and test explanations.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate mean and median for 2,2,2,10. What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The fourth duration remains 10 seconds. Mean is 16/4=4 seconds; median is 2 seconds. Report the timeout **evidence** and distinguish it from an untested explanation.

Changed case: The mean becomes 2 seconds but the selective report violates this plan. Observed inconvenient outcomes are not an approved exclusion reason.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.